

Epidemiology of Alzheimer's disease and other forms of dementia in China, 1990-2010: a systematic review and analysis.

BACKGROUND: China is increasingly facing the challenge of control of the growing burden of non-communicable diseases. We assessed the epidemiology of Alzheimer's disease and other forms of dementia in China between 1990, and 2010, to improve estimates of the burden of disease, analyse time trends, and inform health policy decisions relevant to China's rapidly ageing population. **METHODS:** In our systematic review we searched for reports of Alzheimer's disease or dementia in China, published in Chinese and English between 1990 and 2010. We searched China National Knowledge Infrastructure, Wanfang, and PubMed databases. Two investigators independently assessed case definitions of Alzheimer's disease and dementia: we excluded studies that did not use internationally accepted case definitions. We also excluded reviews and viewpoints, studies with no numerical estimates, and studies not done in mainland China. We used Poisson regression and UN demographic data to estimate the prevalence (in nine age groups), incidence, and standardised mortality ratio of dementia and its subtypes in China in 1990, 2000, and 2010. **FINDINGS:** Our search returned 12,642 reports, of which 89 met the inclusion criteria (75 assessed prevalence, 13 incidence, and nine mortality). In total, the included studies had 340,247 participants, in which 6357 cases of Alzheimer's disease were recorded. 254,367 people were assessed for other forms of dementia, of whom 3543 had vascular dementia, frontotemporal dementia, or Lewy body dementia. In 1990 the prevalence of all forms of dementia was 1·8% (95% CI 0·0-44·4) at 65-69 years, and 42·1% (0·0-88·9) at age 95-99 years. In 2010 prevalence was 2·6% (0·0-28·2) at age 65-69 years and 60·5% (39·7-81·3) at age 95-99 years. The number of people with dementia in China was 3·68 million (95% CI 2·22-5·14) in 1990, 5·62 million (4·42-6·82) in 2000, and 9·19 million (5·92-12·48) in 2010. In the same period, the number of people with Alzheimer's disease was 1·93 million (1·15-2·71) in 1990, 3·71 million (2·84-4·58) people in 2000, and 5·69 million (3·85-7·53) in 2010. The incidence of dementia was 9·87 cases per 1000 person-years, that of Alzheimer's disease was 6·25

cases per 1000 person-years, that of vascular dementia was 2·42 cases per 1000 person-years, and that of other rare forms of dementia was 0·46 cases per 1000 person-years. We retrieved mortality data for 1032 people with dementia and 20,157 healthy controls, who were followed up for 3-7 years. The median standardised mortality ratio was 1·94:1 (IQR 1·74-2·45). INTERPRETATION: Our analysis suggests that previous estimates of dementia burden, based on smaller datasets, might have underestimated the burden of dementia in China. The burden of dementia seems to be increasing faster than is generally assumed by the international health community. Rapid and effective government responses are needed to tackle dementia in low-income and middle-income countries. FUNDING: Nossal Institute of Global Health (University of Melbourne, Australia), the National 12th Five-Year Major Projects of China, National Health and Medical Research Council Australia-China Exchange Fellowship, Importation and Development of High-Calibre Talents Project of Beijing Municipal Institutions, and the Bill & Melinda Gates Foundation. Copyright © 2013 Elsevier Ltd. All rights reserved.